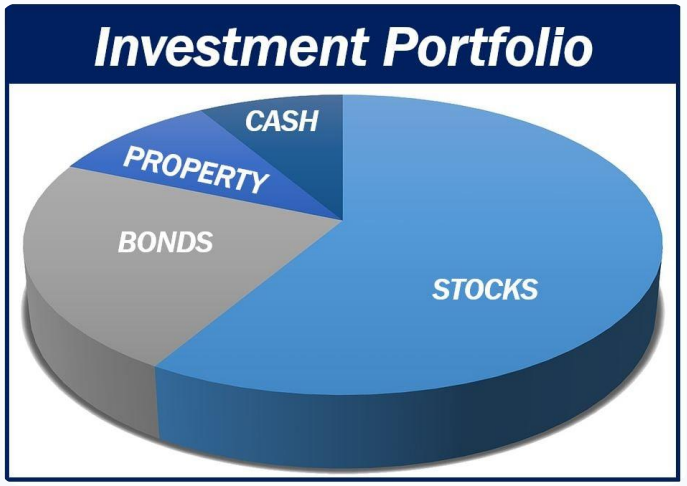


CCBS 4024 Investments and Portfolio Management

https://investmenttopic1-2wzbmason.manus.space/

Topic 2: Introduction to Portfolio return and risk

What is a portfolio (投資組合)?



- Portfolio is a collection of investments or assets held by an institution or a private individual.
- The assets in the portfolio could include stocks, bonds, options, warrants, gold certificates, real estate, futures contracts, production facilities, or any other item that is expected to retain its value.
- The obvious advantage of holding a portfolio is diversification (分散風險) which means “don’t put all eggs in one basket.” in layman’s terms.
- This lecture examines what drives diversification power in a portfolio.

Portfolio's expected return and standard deviation

- Like an individual asset, we can calculate the expected return, $E(R_p)$ and standard deviation, σ_p of a portfolio.

- **The portfolio's expected return** is just the weight average expected return of individual assets in a portfolio. It depends on the weight of each investment you invest and the expected return of each asset in your portfolio.

- **The portfolio's standard deviation** depends on the weight of each investment you invest, the standard deviations of each investment and the **covariance** between investments in your portfolio.



Portfolio weight

- **The portfolio weight of an asset** means the fraction of the total investment held in each individual asset in the portfolio. For example, if we want to find the **weight of an asset i** called $\mathbf{w_i}$ in a portfolio, we can use the following formula to find it.

$$w_i = \frac{market\ value\ of\ investment\ in\ asset\ i}{Total\ market\ value\ of\ portfolio}$$



- The total portfolio weights must add up to 1 (or 100%).
- An individual asset may have a negative weight, indicating that the asset is being **sold short**.

Short selling

1

Short selling an asset means you borrow the asset from others and then sell the asset in the market at the prevailing price to get some cash for other uses now.

2

Usually, investors adopting this strategy expect the price of the asset will decline. To **close out** the short selling position, investors buy back the asset and return it to the lenders later.

3

Short sellers can make profit if the short selling price is higher than buying back price of the asset. If the reverse happens, the short sellers will suffer loss.

Short selling

Table 3.2

Cash flows from purchasing versus short-selling shares of stock

Purchase of Stock		
Time	**Action**	**Cash Flow***
0	Buy share	– Initial price
1	Receive dividend, sell share	Ending price + Dividend
Profit = (Ending price + Dividend) – Initial price		
Short Sale of Stock		
Time	**Action**	**Cash Flow***
0	Borrow share; sell it	+ Initial price
1	Repay dividend and buy share to replace the share originally borrowed	– (Ending price + Dividend)
Profit = Initial price – (Ending price + Dividend)		

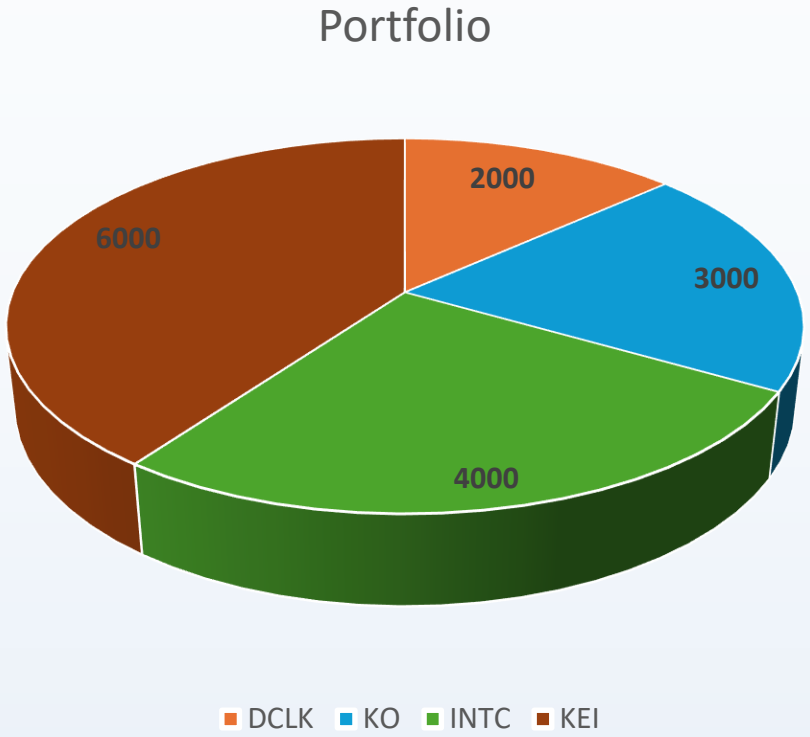
*A negative cash flow implies a cash *outflow*.

Example of calculating the portfolio weight

Suppose you have $15,000 in total to invest in the following 4 securities. Their dollar investments in each security are as follows:

Portfolio:
$2000 of DCLK,
$3000 of KO,
$4000 of INTC, and
$6000 of KEI

Find the weight of each security in the portfolio.



Return of a portfolio

- The return on the portfolio, R_p , is the weighted average of the returns on the investments in the portfolio. Mathematically, if we have n assets in the portfolio, the return of the portfolio is:

$$R_p = w_1 R_1 + w_2 R_2 + w_3 R_3 + \cdots + w_n R_n$$

EXAMPLE

Calculating the portfolio return

THE SCENARIO

Suppose you buy 500 shares of Ford at $11 per share and 100 shares of Citigroup stock at $28 per share. After one month, Ford's share price goes up to $13 and Citigroup's rises to $40. Find the following:

FORD
500 shares
$11 → $13

CITIGROUP
100 shares
$28 → $40

FIND THE FOLLOWING

I. The percentage rate of return of each stock assuming no dividend payments.

II. The portfolio weight of each stock **before** the change in stock prices.

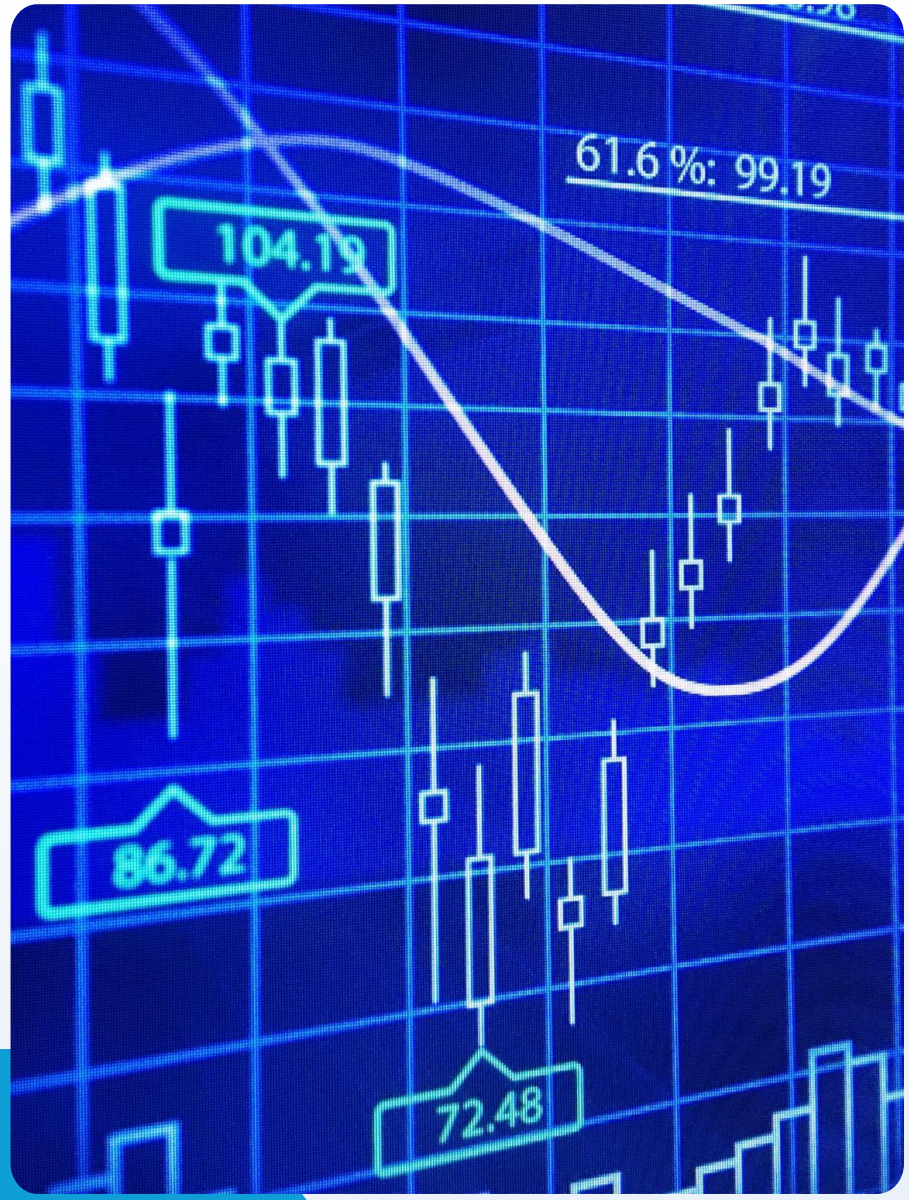
III. The return of the portfolio.

Portfolio Expected Returns

- The **expected return of a portfolio** is the weighted average of the expected returns of the investments within it. Thus, the expected return of a portfolio consisting n assets is calculated as:

- $\mathrm{E}(R_p) = w_1\ E(R_1) + w_2\ E(R_2) + w_3\ E(R_3) + \cdots + w_n\ E(R_n)$





Example – Portfolio expected returns

Assume your portfolio consists of $25,000 of Intel stock and $35,000 of ATP Oil and Gas. Your expected return is 18% for Intel and 25% for ATP Oil and Gas. **What is the expected return for your portfolio?**

Portfolio's standard deviation and Covariance between assets



To find the standard deviation or variance of a portfolio, one must know the degree to which the stocks' returns move together. In statistics, the co-movement of two stocks can be measured by i) **covariance** and ii) **correlation coefficient**.



Covariance is an indicator of the direction of the relationship between two variables. The covariance is unbounded (i.e., ranges from negative infinity to positive infinity) and, therefore, is not a very useful measure of the strength of the relationship between two assets' returns.



If the covariance between two assets is **positive,** the two returns tend to move in the **same directions**. If the covariance is **negative**, the two returns tend to move in **opposite directions**.

Portfolio's standard deviation – Correlation coefficient



- **Correlation coefficient (ρ)** is a measure of the degree of strength of the linear relationship between two variables.
- The correlation between two stocks will always lie between –1 and +1.
- When the correlation is closer to +1, there is a (stronger/weaker) (positive/negative) relationship between two variables.

Different correlation coefficient

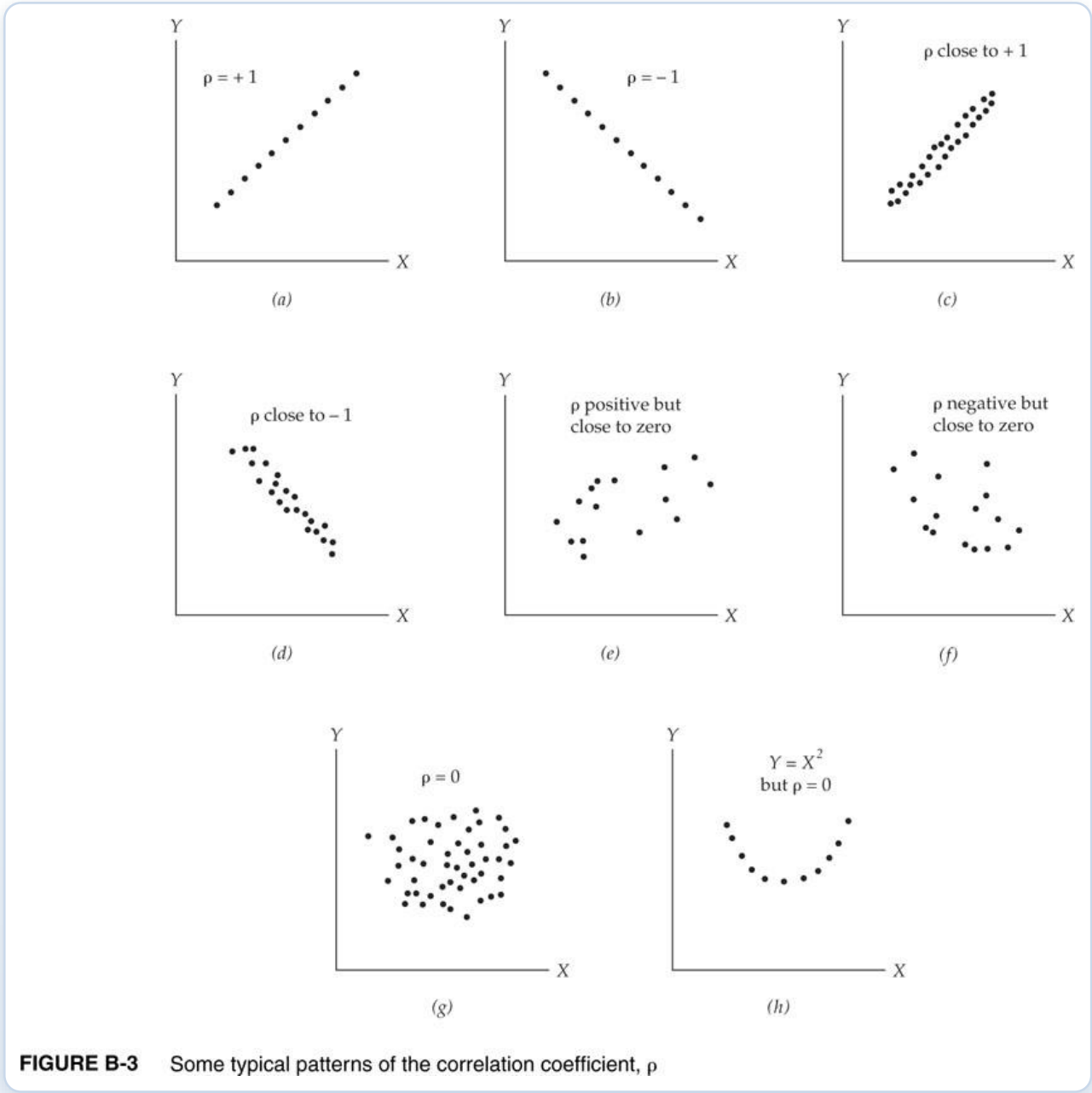


FIGURE B-3 Some typical patterns of the correlation coefficient, ρ

HOW TO READ IT

- **ρ = +1** perfect positive relationship
- **ρ = −1** perfect negative relationship
- **ρ = 0** no linear relationship

Calculating sample covariance and correlation coefficient

- We can estimate of the **sample covariance between 2 assets' return** (asset i and j), $\boldsymbol{\sigma_{ij}}$, from historical data using the formula below.

$$\sigma_{ij} = Cov(R_i, R_j) = \frac{1}{T-1}\sum_t (R_{i,t} - \bar{R}_i)\,(R_{j,t} - \bar{R}_j)$$

- The **correlation coefficient** between 2 assets' return (asset i and j), $\boldsymbol{\rho_{ij}}$, can be calculated by the following formula:

$$\rho_{ij} = Corr(R_i, R_j) = \frac{Cov(R_i, R_j)}{SD(R_i)\,SD(R_j)} \quad \text{or} \quad \sigma_{ij} = \rho_{ij}\sigma_i\,\sigma_j$$

EXAMPLE

Calculating sample covariance and correlation coefficient

YOUR TASK

Find the sample covariance and correlation coefficient between Stock X and Y.

ANNUAL RETURNS · 2018–2022

Year	Stock X	Stock Y
2018	18%	0%
2019	5%	-3%
2020	12%	15%
2021	4%	12%
2022	6%	1%

Calculating covariance and correlation between two assets, stock X and Y

Expected return and variance of a two risky assets' portfolio

- Suppose we have two risky assets, A and B with expected returns, $E(r_A)$ and $E(r_B)$ respectively and with S.D., σ_A and σ_B respectively.

- Portfolio expected return = $E(r_P) = w_A E(r_A) + w_B E(r_B)$

- Portfolio variance, $\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B Cov(r_A, r_B)$

  or $$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + 2w_A w_B Corr(r_A, r_B)\sigma_A\sigma_B$$

- Portfolio standard deviation is still the square root of the portfolio variance.



Example: Calculating the portfolio's expected returns and standard deviations

- **Use the information in Silde 16 to answer this question.**
- Suppose you invest 60% of your money in Stock X and 40% in Stock Y. Find:

1. the expected return of the portfolio and;
2. the standard deviation of return of the portfolio.

Answer

Example : Calculating the portfolio's expected returns and standard deviations

Using the information in the table, calculate the expected return and standard deviation of the 2-asset (debt and equity) portfolio if you invest $40,000 in debt and $60,000 in equity.

	Debt	Equity
Expected return, $E(r)$	8%	13%
Standard deviation, σ	12%	20%
Covariance, $Cov(r_D, r_E)$		72
Correlation coefficient, ρ_{DE}	0.00	.30

TABLE 7.1

Descriptive statistics for two mutual funds

Case study: Bond and Stock

w_D	w_E	$E(r_P)$	Portfolio Standard Deviation for Given Correlation $\rho = -1$	$\rho = 0$	$\rho = .30$	$\rho = 1$
0.00	1.00	13.00	20.00	20.00	20.00	20.00
0.10	0.90	12.50	16.80	18.04	18.40	19.20
0.20	0.80	12.00	13.60	16.18	16.88	18.40
0.30	0.70	11.50	10.40	14.46	15.47	17.60
0.40	0.60	11.00	7.20	12.92	14.20	16.80
0.50	0.50	10.50	4.00	11.66	13.11	16.00
0.60	0.40	10.00	0.80	10.76	12.26	15.20
0.70	0.30	9.50	2.40	10.32	11.70	14.40
0.80	0.20	9.00	5.60	10.40	11.45	13.60
0.90	0.10	8.50	8.80	10.98	11.56	12.80
1.00	0.00	8.00	12.00	12.00	12.00	12.00
			Minimum Variance Portfolio			
		w_D	0.6250	0.7353	0.8200	—
		w_E	0.3750	0.2647	0.1800	—
		$E(r_P)$	9.8750	9.3235	8.9000	—
		σ_P	0.0000	10.2899	11.4473	—

TABLE 7.3

Expected return and standard deviation with various correlation coefficients



Why do we invest in a portfolio?

- **Diversification** means the practice of spreading investments across different asset classes, sectors, and geographic regions in order to reduce risk.
- The idea is that the loss of an asset may be compensated by the gain of another asset. Then portfolio's return will be (less / more) volatile than that of an individual asset.
- But how can we maximize the benefit of diversification?

The Principle of Diversification

- Diversification can be achieved through:

i) simply holding larger number of assets (naïve diversification) and;

ii) selecting assets with low correlation coefficient.

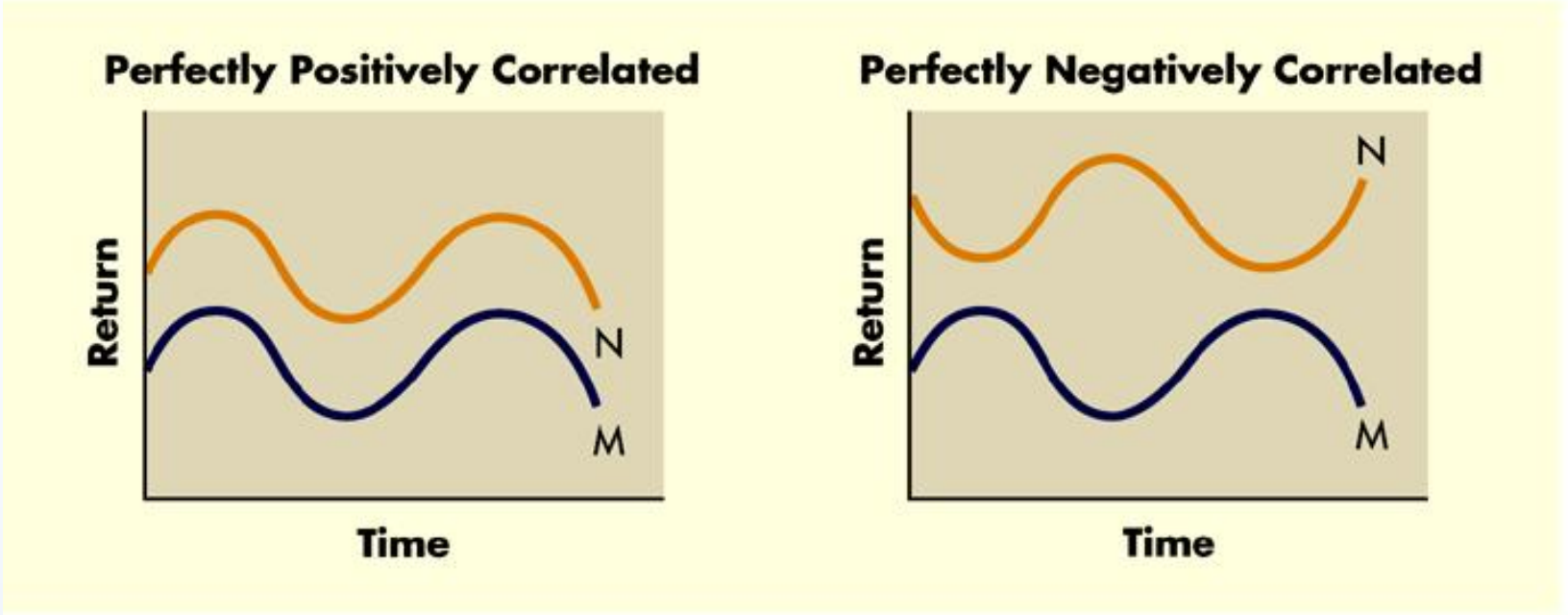
- If we just blindly hold more and more assets in the portfolio without studying the correlation coefficient between assets, it is called **naïve diversification**. But this method of diversification may not be effective in reducing risk. For example, if an investor may hold a lot of stocks (China Construction Bank, Bank of China, Ping On Insurance Group of China, … etc.) with similar characteristics, the portfolio is not well-diversified.



The Principle of Diversification

- A cleverer way to achieve diversification is to select stocks with lower correlation coefficient even if the portfolio has 2 assets only.
- If the correlation is perfectly positive (+1), there is no gain from diversification.
- If the correlation is between +1 and – 1, there are some benefits of diversification.
- If the correlation is perfectly negative (–1), diversification provides the greatest potential risk reduction. In this scenario, it is possible to construct a portfolio with zero risk.

Diagram

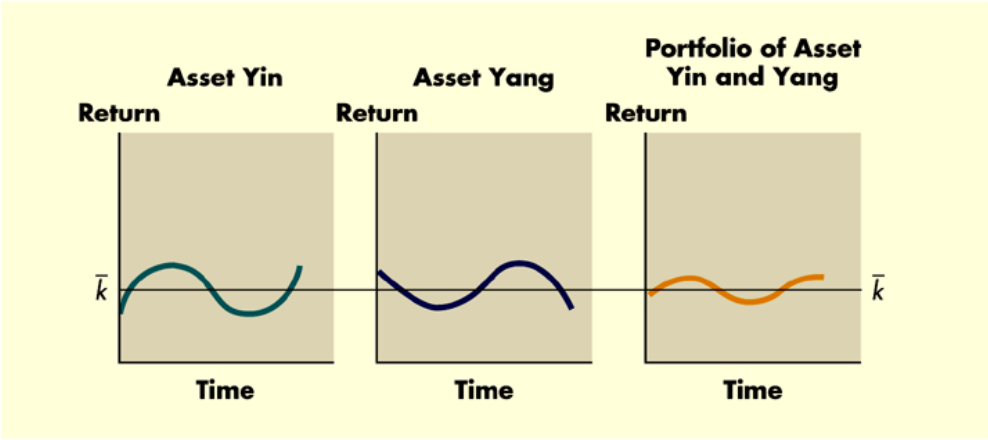


Correlation coefficient =1

Correlation coefficient =-1

Case: -1<ρ< 1

- Even if two assets are not perfectly negatively correlated (the correlation coefficient is between -1 and 1), an investor can still realize diversification benefits from combining them in a portfolio.



Types of risk

1. **Unique Risk/ Firm-specific risk/ Idiosyncratic risk/ Diversifiable risk (可分散風險)/ Unsystematic risk/ Residual risk/ Independent risk:**

   This is the risk that can be potentially fully eliminated by diversification. It refers to the fact that the risk surrounding an individual company are peculiar to that company only and hence can be eliminated through a well-diversified portfolio.

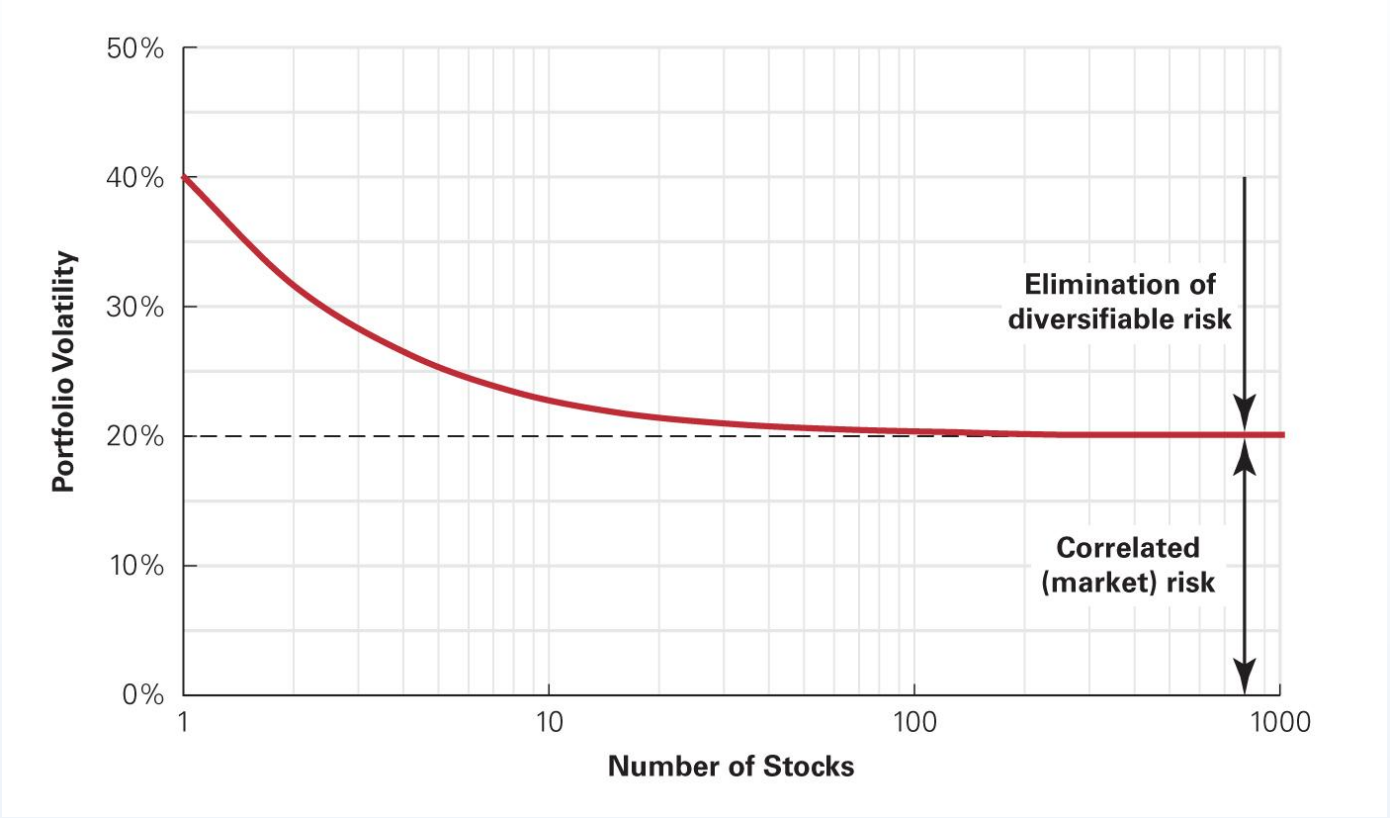
2. **Market Risk (市場風險)/ Systematic risk (系統性風險)/ Undiversifiable risk/ non-diversifiable risk/ Beta risk/ Common risk:**

   The risk is unavoidable. Regardless of how many stocks or assets you choose in your portfolio (how much you diversify), market risk is still there. It stems from the fact that there are many economy-wide perils that threaten ***all*** businesses. No matter how many stocks you hold, you'd still be ***exposed*** to certain market uncertainties.

Types of risk

3. **Total risk** (standard deviation) of a portfolio is the sum of diversifiable risk and non-diversifiable risk.

Total risk = diversifiable risk + non-diversifiable risk



Types of risk

Exercise

Diversifiable Versus Systematic Risk

Which of the following risks of a stock are likely to be firm-specific, diversifiable risks, and which are likely to be systematic risks? Which risks will affect the risk premium that investors will demand?

a. The risk that the founder and CEO retires
b. The risk that oil prices rise, increasing production costs
c. The risk that a product design is faulty, and the product must be recalled
d. The risk that the economy slows, reducing demand for the firm's products

Solution

- Because oil prices and the health of the economy affect all stocks, risks (b) and (d) are systematic risks. These risks are not diversified in a large portfolio, and so will affect the premium that investors require to invest in a stock.

- Risks (a) and (c) are firm-specific risks, and so are diversifiable. While these risks should be considered when estimating a firm's future cash flows, they will not affect the risk premium that investors will require and, therefore, will not affect a firm's cost of capital.

CONCEPT

Portfolios of one risky asset and a risk-free asset

Three things to know about the risk-free asset

1

DEFINING FEATURE

A risk-free asset has a certain payoff. Its standard deviation is zero, so its correlation with any risky asset is also zero.

2

REAL-WORLD EXAMPLE

In finance, a commonly used example of a risk-free asset is the U.S. Treasury bills.

3

WHY IT MATTERS

Expected return and standard deviation for a portfolio of one risky asset and a risk-free asset are simpler to calculate than for two risky assets.

Portfolios of one risky asset and a risk-free asset

- Suppose w_A is the weight you invest in the risky Asset A and thus w_{R_F} or $(1 - w_A)$ is the weight you invest in the risk-free asset.

1. The expected return on the portfolio consisting Asset A and the risk-free asset is:

$$E(R_p)=w_A E(R_A) + (w_{R_F})R_F = w_A\ E(R_A) + (1 - w_A)R_F$$

2. The portfolio's standard deviation = $\sigma_p = w_A \sigma_A$

- When w_{R_F} is negative, it means short-selling the risk-free asset or simply **borrowing**.
- A **complete portfolio** is defined as a combination of a risky asset portfolio, with return Rp, and the risk-free asset, with return R_F.

Example

You manage a risky portfolio with expected return of 18% and standard deviation of 28%. The T-bill rate is 8%. Your client chooses to invest 70% of a portfolio in your fund and 30% in a risk-free T-bill money market fund. What is the expected value and standard deviation of the rate of return on your client's portfolio?

E(R) and S.D. of a portfolio with three or more risky assets

- Suppose we form a portfolio using **three risky assets, A, B and C**

➢ The expected return on three-asset portfolio = $E(R_p) = w_A E(r_A) + w_B E(r_B) + w_C E(r_C)$

➢ The three-asset portfolio's variance:

$$\sigma_p^2 = w_A^2\sigma_A^2 + w_B^2\sigma_B^2 + w_C^2\sigma_C^2 + 2w_A w_B Cov(r_A, r_B) + 2w_A w_C Cov(r_A, r_C) + 2w_B w_C Cov(r_B, r_C)$$

- In general, for **n-risky asset portfolios**,

$$E(R_p) = \sum_i w_i E(R_i)$$

$$\sigma_p^2 = \sum_{i=1}^{n}\sum_{j=1}^{n} w_i w_j \mathrm{Cov}\left(r_i, r_j\right)$$

Self-study: Markowitz Portfolio Optimization Model

- Recall for n assets:

$$\sigma_p^2 = \sum_{i=1}^{n} \sum_{j=1}^{n} w_i w_j \mathrm{Cov}\left(r_i, r_j\right)$$

- Define the average variance, $\overline{\sigma^2}$ and average covariance, $\overline{Cov}$ of the securities as:

$$\overline{\sigma^2} = \frac{1}{n} \sum_{i=1}^{n} \sigma_i^2$$

$$\overline{Cov} = \frac{1}{n(n-1)} \sum_{\substack{j=1 \\ j \neq i}}^{n} \sum_{i=1}^{n} \mathrm{Cov}(r_i, r_j)$$

Self-study: Markowitz Portfolio Optimization Model

- Assume **each asset has an equal weight** in the portfolio. We can then express portfolio variance in terms of the average variance and the average covariance as:

$$\sigma_p^2 = \frac{1}{n}\bar{\sigma}^2 + \frac{n-1}{n}\overline{Cov}$$

- Portfolio variance can be smaller when the number of assets (n) rises initially. However, the portfolio variance won't usually fall to 0 even an investor buys a lot of assets in the portfolio.
- The risk of a highly diversified portfolio depends on the covariance of the returns of the component securities.